# PulseIQ Analytics | Phase 3

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**GitHub:** PoonamPatole/pulseiq
**Git tag:** Phase-3

## Task status and evidence

| Task | Status | Evidence |
|---|---|---|
| 11. P&L and expenses | Complete locally; P&L ties to governed M16-M18 views. | Findings and tie-out |
| 12. Receivables and cash flow | Complete locally; ageing, payment timing, and cash measures built. | Bucket check |
| 13. Cross-department impact | Complete locally; 13 shared measures reconcile to PQ-13. | Comparison notes |
| 14. AI signals and forecasts | Complete locally; anomalies and forecasts include uncertainty notes. | Forecast notes |
| 15. Executive scorecard | Complete locally; eight governed KPIs and supplied targets. | KPI selection |

## Proof it works

**Task 11:** Warehouse counts match the source: 6,784 ledger and 501 expense rows. Revenue, net profit, and margin tie to M16-M18 for April 2025-September 2026. From March to April 2026, revenue rose 11.4%, expenses 20.7%, and profit fell from INR 15.48m to INR 11.45m.

**Task 12:** Receivables use four non-overlapping age bands; their totals check against open balances. Cash-in and cash-out follow payment dates, with GST included for paid bills.

**Task 13:** All 2,574 checks (198 department-months x 13 measures) match PQ-13 after eight documented exclusions. Source records remain unchanged.

**Task 14:** 405 anomalies include INR 31.42m estimated payroll exposure across 184 records; this is for review, not confirmed loss. Twelve forecasts have no actuals yet, so error and MAPE are unavailable.

**Task 15:** The scorecard has eight governed KPIs. Targets are used only when supplied; four KPIs have no numeric target.

## What I learned

I learned to keep financial measures tied to their source and date basis, aggregate facts to a shared grain before joining, and present model signals as uncertain. Targets should come from business owners, not inference.

## Problems and how I handled them

The workbook headlines are counts, so Task 11 ties to governed metrics. Task 13 documents eight excluded records and preserves source facts. One approved leave request has reversed dates and is assigned to its start month for review. Task 14 has no actuals in its forecast horizon; blank anomaly estimates are not treated as zero. Task 15 leaves four KPIs without supplied targets unscored.

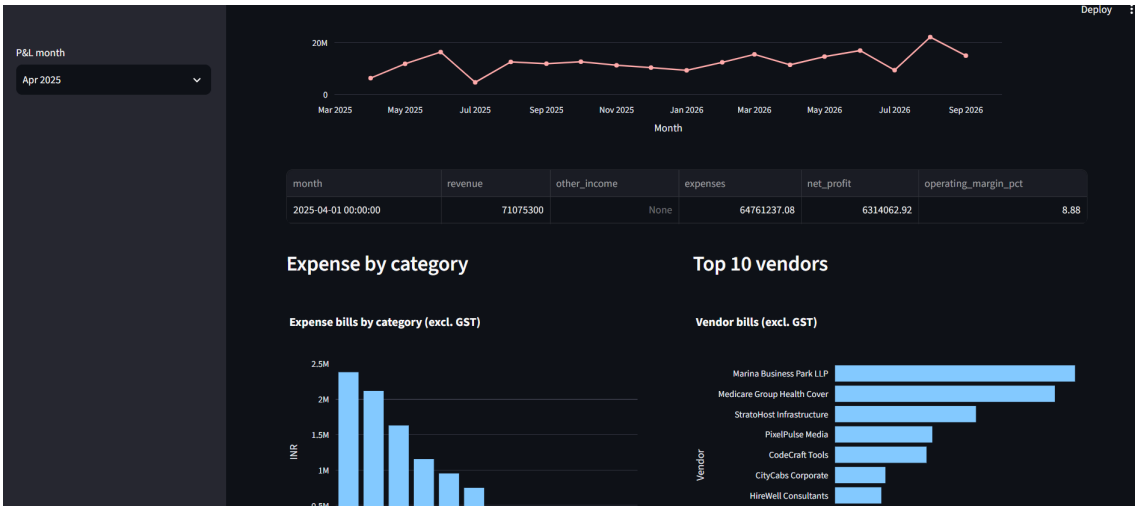
## Plan for the next phase

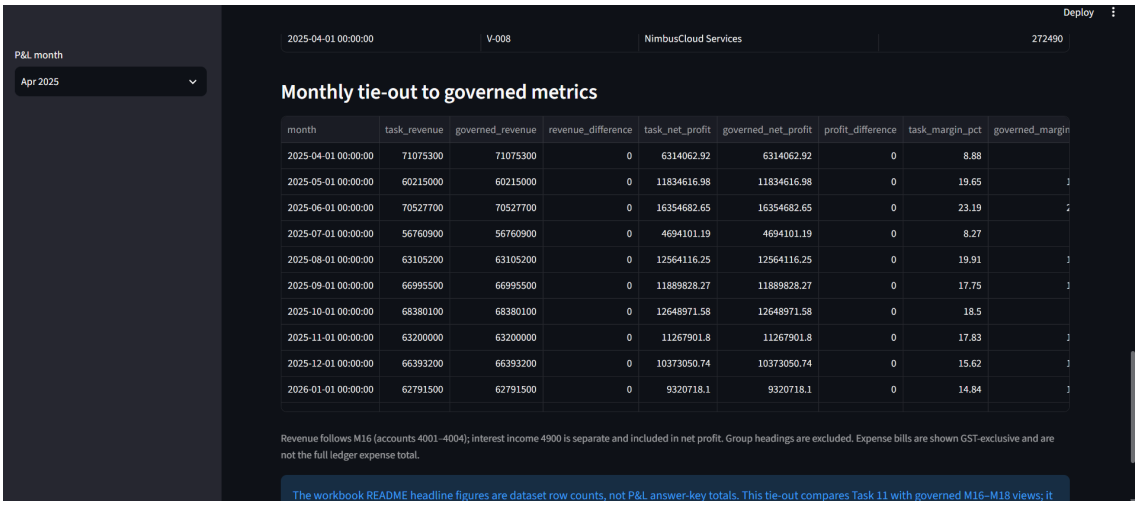
Review payroll and source exceptions with their owners, compare forecasts when actuals arrive, confirm KPI targets, and package the dashboards and evidence for final submission.

# Task screenshots

## Task 11. P&L and expense analytics



# Task 12. Invoices, receivables, and cash flow

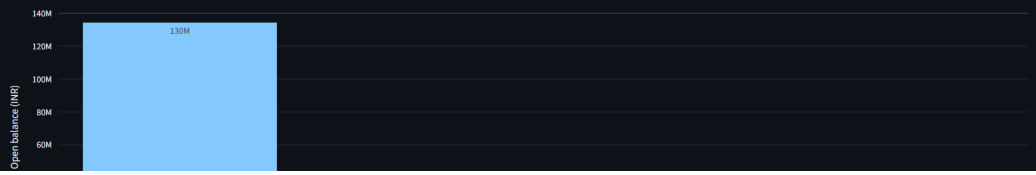
## Task 12 | Invoices, receivables, and cash flow

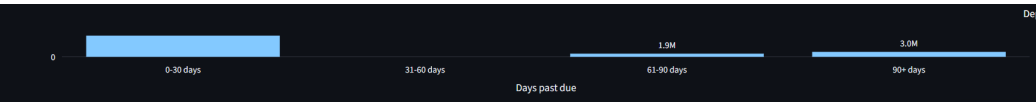
**Open invoice statuses:** SENT, OVERDUE, and PARTIALLY_PAID. DRAFT, PAID, and CANCELLED are excluded.

Ageing uses days past due at the dataset cutoff. The 90+ bucket means more than 90 days, keeping the buckets exclusive. Zero days past due is included in 0–30 days.

| Open receivables | Open invoices | Bucket total difference |
|---|---|---|
| ₹139,394,278 | 62 | ₹0.00 |

### Receivables ageing




| ageing_bucket | invoice_count | receivables |
|---|---|---|
| 0-30 days | 59 | 134417038 |
| 31-60 days | 0 | 0 |
| 61-90 days | 1 | 1940982 |
| 90+ days | 2 | 3036258 |

Bucket check: ₹139,394,278.00 across the four buckets equals open receivables of ₹139,394,278.00 (difference ₹0.00).

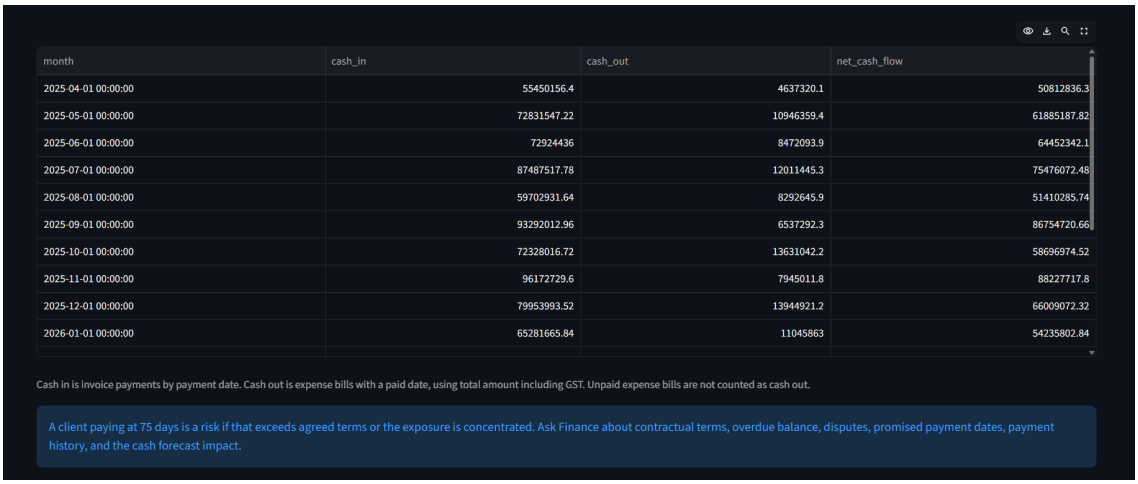
### Clients with the most overdue money

| client_id | client_name | payment_terms_days | overdue_invoice_count | overdue_balance | oldest_days_past_due |
|---|---|---|---|---|---|
| CL-015 | Marina Media Works | 30 | 2 | 7110208 | 11 |
| CL-014 | Siruvani Agro Tech | 45 | 3 | 6097296 | 548 |
| CL-004 | Kaveri Textiles Pvt Ltd | 60 | 1 | 3884442 | 12 |
| CL-019 | Nordic Freight AB | 30 | 2 | 3632100 | 6 |
| CL-010 | PeakEdge Education Pvt Ltd | 30 | 1 | 1094096 | 3 |
| CL-021 | Tasman Retail Pty Ltd | 30 | 1 | 969912 | 7 |
| CL-017 | Greenfield Realty Ltd | 45 | 1 | 968662 | 2 |

### Average days to pay by client

| client_id | client_name | payment_terms_days | paid_invoice_count | average_days_to_pay | maximum_days_to_pay |
|---|---|---|---|---|---|
| CL-016 | Triveni Steel Industries | 60 | 12 | 121.2 | 180 |
| CL-014 | Siruvani Agro Tech | 45 | 11 | 105.4 | 145 |
| CL-004 | Kaveri Textiles Pvt Ltd | 60 | 40 | 95.6 | 149 |
| CL-025 | Gulf Mercantile LLC | 60 | 8 | 91.5 | 108 |
| CL-011 | Zephyr Hospitality Group | 60 | 25 | 89.6 | 115 |
| CL-007 | Vaigai Foods & Beverages | 45 | 33 | 77.4 | 124 |
| CL-012 | Indus Power Utilities | 60 | 14 | 71.1 | 88 |
| CL-022 | Rheinwerk Systems GmbH | 45 | 16 | 57.1 | 91 |
| CL-008 | Coromandel Auto Components | 45 | 25 | 56.5 | 73 |
| CL-006 | NorthStar Telecom Ltd | 45 | 42 | 54.5 | 94 |



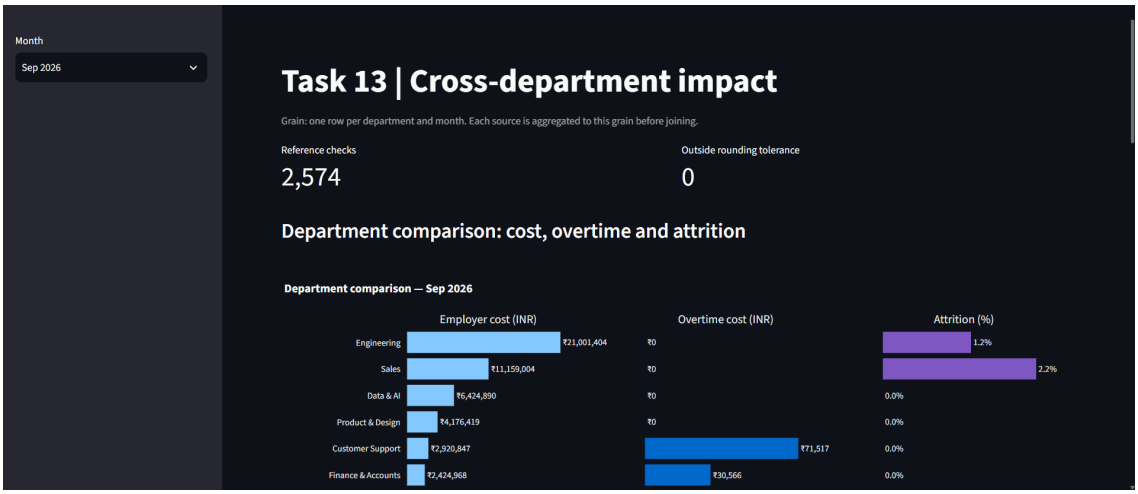



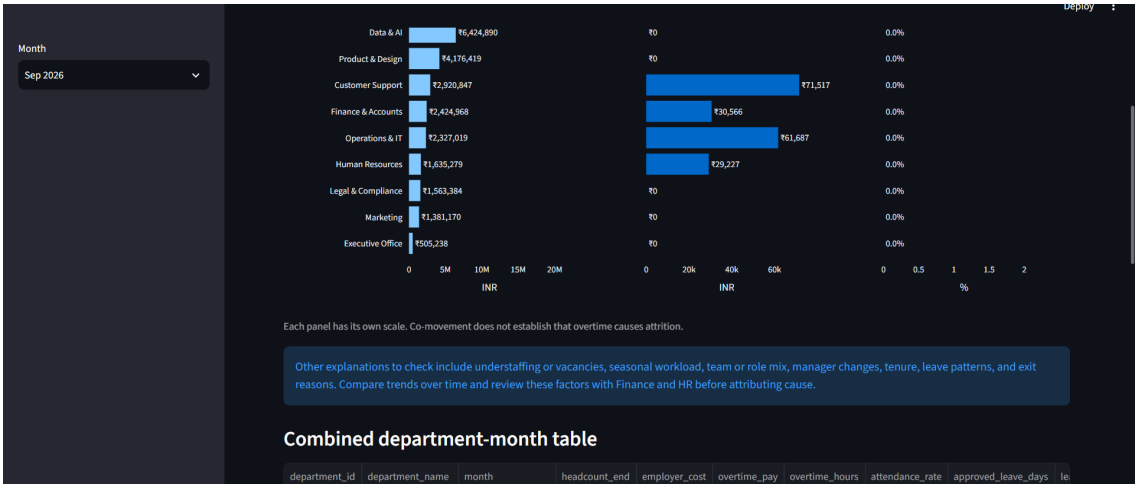
| month | cash_in | cash_out | net_cash_flow |
|---|---|---|---|
| 2025-04-01 00:00:00 | 55450156.4 | 4637320.1 | 50812836.3 |
| 2025-05-01 00:00:00 | 72831547.22 | 10946359.4 | 61885187.82 |
| 2025-06-01 00:00:00 | 72924436 | 8472093.9 | 64452342.1 |
| 2025-07-01 00:00:00 | 87487517.78 | 12011445.3 | 75476072.48 |
| 2025-08-01 00:00:00 | 59702931.64 | 8292645.9 | 51410285.74 |
| 2025-09-01 00:00:00 | 93292012.96 | 6537292.3 | 86754720.66 |
| 2025-10-01 00:00:00 | 72328016.72 | 13631042.2 | 58696974.52 |
| 2025-11-01 00:00:00 | 96172729.6 | 7945011.8 | 88227717.8 |
| 2025-12-01 00:00:00 | 79953993.52 | 13944921.2 | 66009072.32 |
| 2026-01-01 00:00:00 | 65281665.84 | 11045863 | 54235802.84 |

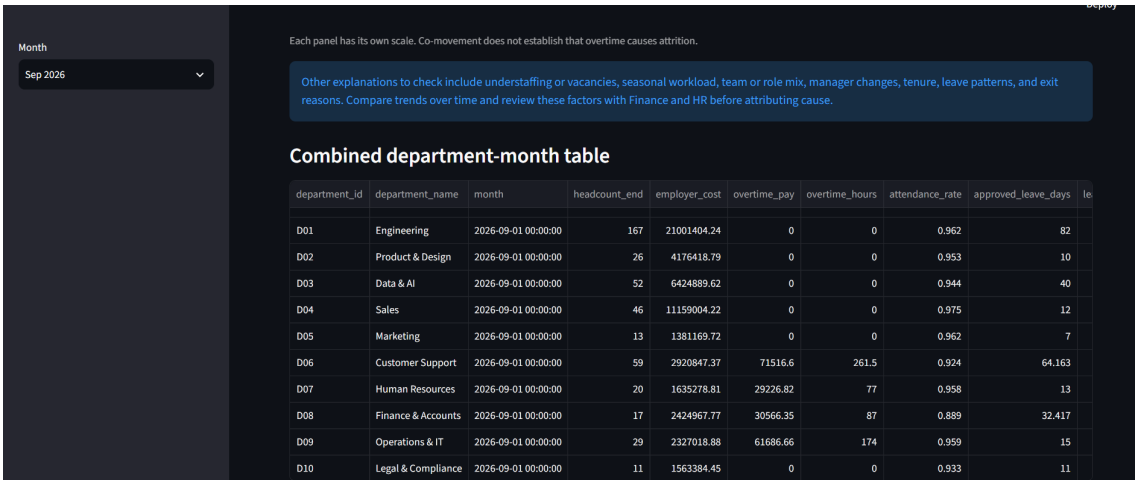
Cash in is invoice payments by payment date. Cash out is expense bills with a paid date, using total amount including GST. Unpaid expense bills are not counted as cash out.

A client paying at 75 days is a risk if that exceeds agreed terms or the exposure is concentrated. Ask Finance about contractual terms, overdue balance, disputes, promised payment dates, payment history, and the cash forecast impact.

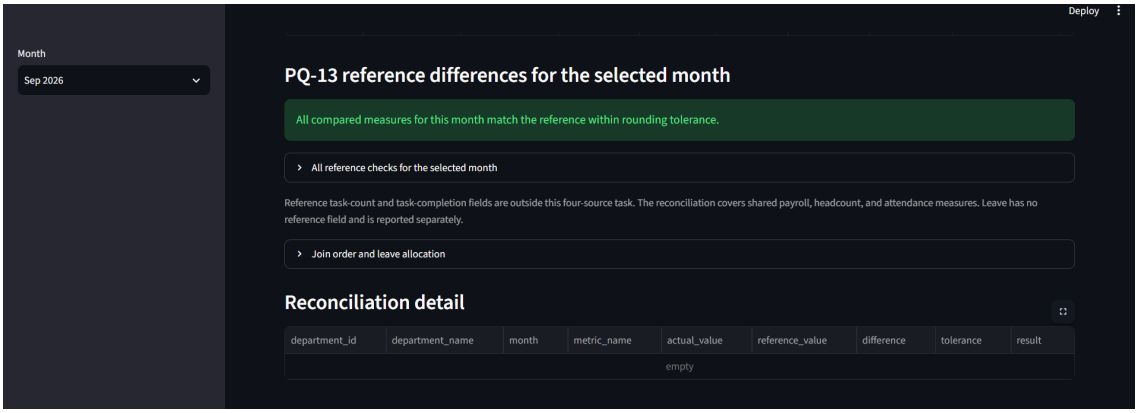
## Task 13. Cross-department impact





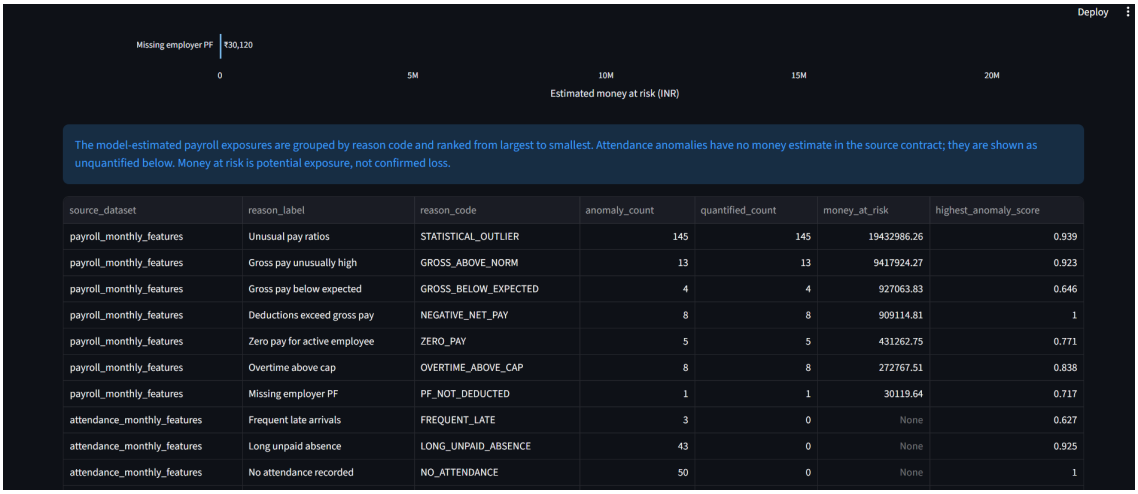




| department_id | department_name | month | headcount_end | employer_cost | overtime_pay | overtime_hours | attendance_rate | approved_leave_days | le |
|---|---|---|---|---|---|---|---|---|---|
| D01 | Engineering | 2026-09-01 00:00:00 | 167 | 21001404.24 | 0 | 0 | 0.962 | 82 | |
| D02 | Product & Design | 2026-09-01 00:00:00 | 26 | 4176418.79 | 0 | 0 | 0.953 | 10 | |
| D03 | Data & AI | 2026-09-01 00:00:00 | 52 | 6424889.62 | 0 | 0 | 0.944 | 40 | |
| D04 | Sales | 2026-09-01 00:00:00 | 46 | 1115904.22 | 0 | 0 | 0.975 | 12 | |
| D05 | Marketing | 2026-09-01 00:00:00 | 13 | 1381169.72 | 0 | 0 | 0.962 | 7 | |
| D06 | Customer Support | 2026-09-01 00:00:00 | 59 | 2920847.37 | 71516.6 | 261.5 | 0.924 | 64.163 | |
| D07 | Human Resources | 2026-09-01 00:00:00 | 20 | 1635278.81 | 29226.82 | 77 | 0.958 | 13 | |
| D08 | Finance & Accounts | 2026-09-01 00:00:00 | 17 | 2424967.77 | 30566.35 | 87 | 0.889 | 32.417 | |
| D09 | Operations & IT | 2026-09-01 00:00:00 | 29 | 2327018.88 | 61686.66 | 174 | 0.959 | 15 | |
| D10 | Legal & Compliance | 2026-09-01 00:00:00 | 11 | 1563384.45 | 0 | 0 | 0.933 | 11 | |

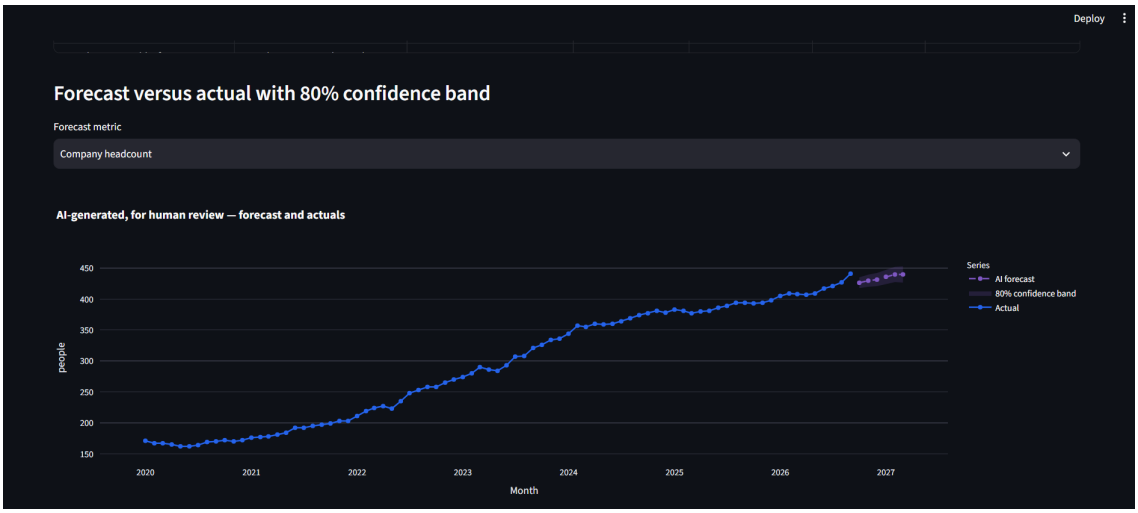


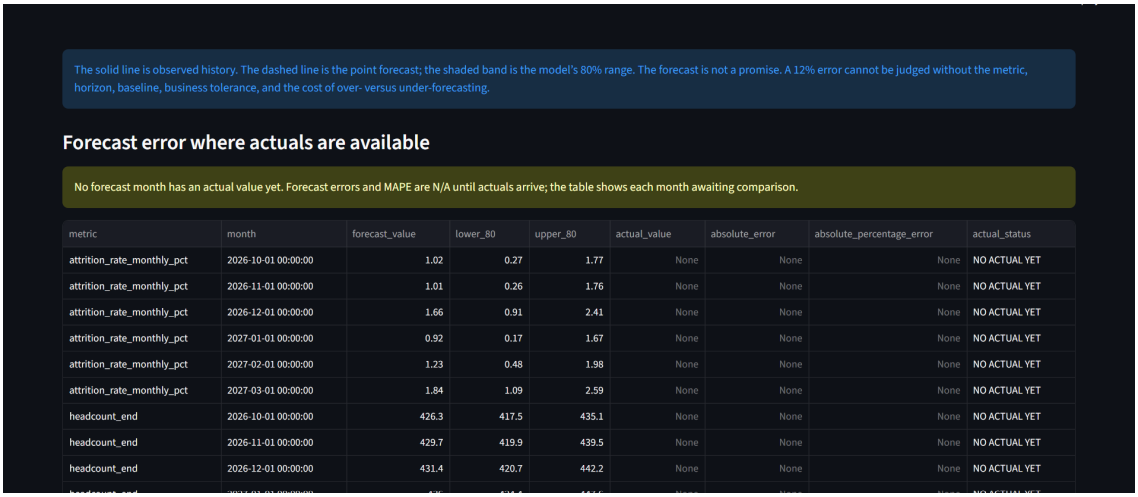
# Task 14. AI signals and forecasts




| source_dataset | reason_label | reason_code | anomaly_count | quantified_count | money_at_risk | highest_anomaly_score |
|---|---|---|---|---|---|---|
| payroll_monthly_features | Unusual pay ratios | STATISTICAL_OUTLIER | 145 | 145 | 19432986.26 | 0.939 |
| payroll_monthly_features | Gross pay unusually high | GROSS_ABOVE_NORM | 13 | 13 | 9417924.27 | 0.923 |
| payroll_monthly_features | Gross pay below expected | GROSS_BELOW_EXPECTED | 4 | 4 | 927063.83 | 0.646 |
| payroll_monthly_features | Deductions exceed gross pay | NEGATIVE_NET_PAY | 8 | 8 | 909114.81 | 1 |
| payroll_monthly_features | Zero pay for active employee | ZERO_PAY | 5 | 5 | 431262.75 | 0.771 |
| payroll_monthly_features | Overtime above cap | OVERTIME_ABOVE_CAP | 8 | 8 | 272767.51 | 0.838 |
| payroll_monthly_features | Missing employer PF | PF_NOT_DEDUCTED | 1 | 1 | 30119.64 | 0.717 |
| attendance_monthly_features | Frequent late arrivals | FREQUENT_LATE | 3 | 0 | None | 0.627 |
| attendance_monthly_features | Long unpaid absence | LONG_UNPAID_ABSENCE | 43 | 0 | None | 0.925 |
| attendance_monthly_features | No attendance recorded | NO_ATTENDANCE | 50 | 0 | None | 1 |






| metric | month | forecast_value | lower_80 | upper_80 | actual_value | absolute_error | absolute_percentage_error | actual_status |
|---|---|---|---|---|---|---|---|---|
| attrition_rate_monthly_pct | 2026-10-01 00:00:00 | 1.02 | 0.27 | 1.77 | None | None | None | NO ACTUAL YET |
| attrition_rate_monthly_pct | 2026-11-01 00:00:00 | 1.01 | 0.26 | 1.76 | None | None | None | NO ACTUAL YET |
| attrition_rate_monthly_pct | 2026-12-01 00:00:00 | 1.66 | 0.91 | 2.41 | None | None | None | NO ACTUAL YET |
| attrition_rate_monthly_pct | 2027-01-01 00:00:00 | 0.92 | 0.17 | 1.67 | None | None | None | NO ACTUAL YET |
| attrition_rate_monthly_pct | 2027-02-01 00:00:00 | 1.23 | 0.48 | 1.98 | None | None | None | NO ACTUAL YET |
| attrition_rate_monthly_pct | 2027-03-01 00:00:00 | 1.84 | 1.09 | 2.59 | None | None | None | NO ACTUAL YET |
| headcount_end | 2026-10-01 00:00:00 | 426.3 | 417.5 | 435.1 | None | None | None | NO ACTUAL YET |
| headcount_end | 2026-11-01 00:00:00 | 429.7 | 419.9 | 439.5 | None | None | None | NO ACTUAL YET |
| headcount_end | 2026-12-01 00:00:00 | 431.4 | 420.7 | 442.2 | None | None | None | NO ACTUAL YET |

## Task 15. Executive company-health scorecard

